DHVANI — Pins & Values Roadmap

Every pin, every part value, in build order · STM32U575ZI-Q + DAC0800 + TL074 on a breadboard

This is the wiring reference. It replaces the wiring pages of the Breadboard Build v2 PDF and includes every change made since: 100 nF on DAC pin 16, a 3 kΩ feedback resistor instead of 2.7 kΩ, and the rail layout. The firmware (DMA, chirps, sensors) is unchanged — keep using v2 for that.

The build order at a glance

Step	What	Done when
1	Rails: label, join the two GNDs	Top blue beeps to bottom blue
2	Nucleo power wires (5V, GND)	Top red reads ~5 V with USB in
3	Battery → −9V rail	Bottom red reads −9 V. Then unclip it
4	DAC0800: parts, then wires	All 16 pins done, beep checks pass
5	TL074: power, amp A, parking	All 14 pins done, beep checks pass
6	8 data wires to the Nucleo	Neighbouring data pins don't beep
7	Power up, flash the DC test	**Gate 2:** 0 / 1.5 / 3.0 V at TL074 pin 1
8	Hour 19: filter + buffer (amps B, C)	Smooth sine on the scope

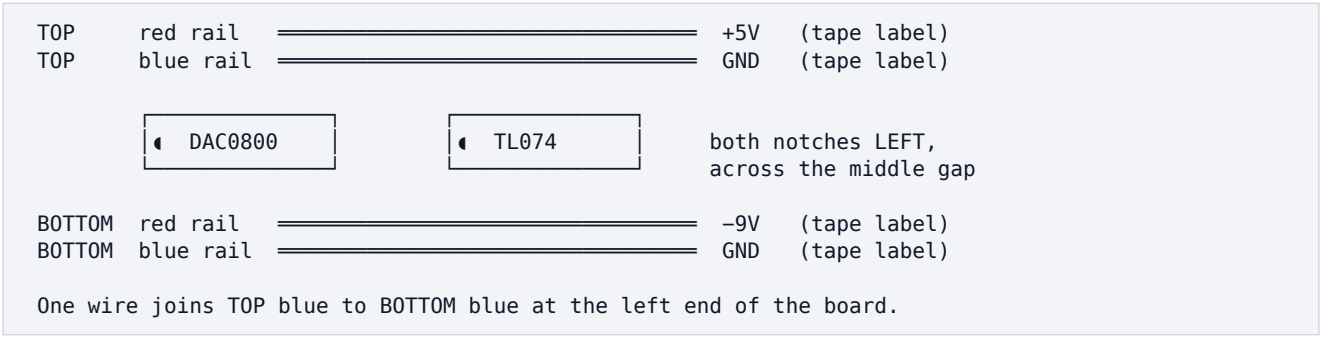
1 · PARTS AND THEIR MARKINGS

Part	Marking	Qty	Where
100 nF ceramic capacitor	104	5	DAC pins 3, 13, 16 · TL074 pins 4, 11
3.3 kΩ resistor	orange-orange-red	2	DAC pins 14, 15
3 kΩ resistor	orange-black-red	1	TL074 pin 1 ↔ pin 2 (feedback)
22 pF ceramic capacitor (optional)	22	1	Across the 3 kΩ
1.5 kΩ resistor	brown-green-red	2	Filter, hour 19 — keep these free
150 pF ceramic capacitor	151	2	Filter, hour 19
51 Ω, 1 kΩ	green-brown-black, brown-black-red	1 each	Scope output, buzzer tap (hour 19)
DAC0800 (16-pin DIP)	—	1	The DAC
TL074 (14-pin DIP)	—	1	All op-amp stages
9 V battery + clip	—	1	−9V rail
Passive piezo buzzer	—	1	Audible demo

> **Ceramic capacitors have no polarity.** Either leg can go either way.
>
> **Resistor colour codes** read from the band nearest one end; the last (gold) band is tolerance and can be ignored.

2 · BREADBOARD BASICS AND LAYOUT

Columns: each chip leg sits in a short column of 5 holes that are joined underneath. To connect anything to a pin, plug it into a free hole in that pin's column — never into the chip. **Rails:** the long strips along the top and bottom edges run the full length of the board.

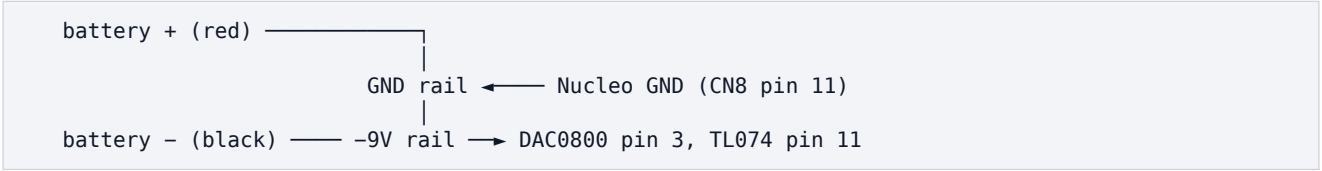


- **Label every rail with tape.** Mixing up +5V and −9V is the one mistake that can kill a chip.
- **Split rails:** a gap in the red/blue line means the rail is broken in the middle — bridge it with a short wire.
- **Pin numbering, notch left:** pin 1 is bottom-left. Count along the bottom row to the right, then back along the top row to the left.
- Place the TL074 right next to the DAC0800 so the DAC pin 4 → TL074 pin 2 wire is only a few centimetres.

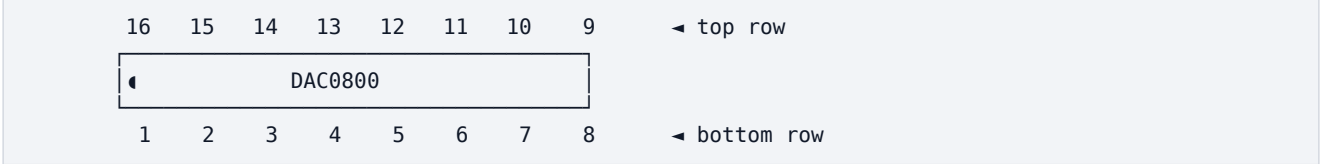
3 · POWER

From	To	Notes
Nucleo CN8 pin 11 (GND)	Top blue rail	Nucleo ground = circuit ground
Nucleo CN8 pin 9 (5V)	Top red rail (+5V)	Feeds DAC pin 13 and TL074 pin 4
Nucleo CN8 pin 7 (3V3)	An empty column → 3.3 kΩ → DAC pin 14	Not a rail — one wire, one resistor
Battery clip **red** (+)	**Bottom blue rail (GND)**	Yes — red goes to GND
Battery clip **black** (−)	Bottom red rail (now −9V)	Feeds DAC pin 3 and TL074 pin 11
Top blue rail	Bottom blue rail	One wire, joins the two GNDs

Why red goes to GND: the battery's + terminal is 9 V above its − terminal. Tie + to GND (0 V) and − has to sit 9 V below it — that is −9 V. All the GNDs — battery red, Nucleo GND, every "to GND" below — are one and the same rail.



4 · DAC0800 — ALL 16 PINS



Pin	Name	Connect to	Part
1	VLC	GND rail	Wire
2	IO-bar	GND rail	Wire
3	V−	Bottom red rail (−9V) + 100 nF to GND	Wire + 104
4	IO (output)	TL074 pin 2	Short wire
5	B1 — MSB	Nucleo PE7 · CN10 pin 20	Wire
6	B2	Nucleo PE6 · CN9 pin 20	Wire
7	B3	Nucleo PE5 · CN9 pin 18	Wire
8	B4	Nucleo PE4 · CN9 pin 16	Wire
9	B5	Nucleo PE3 · CN9 pin 22	Wire
10	B6	Nucleo PE2 · CN9 pin 14	Wire
11	B7	Nucleo PE1 · **CN11 pin 61** (morpho header)	Wire
12	B8 — LSB	Nucleo PE0 · CN10 pin 33	Wire
13	V+	Top red rail (+5V) + 100 nF to GND	Wire + 104
14	VREF+	3.3 kΩ in series → empty column ← wire from CN8 pin 7 (3V3)	3.3 kΩ
15	VREF−	3.3 kΩ to GND rail	3.3 kΩ
16	COMP	100 nF to **pin 3** (not to GND)	104

What each column holds



- **Side branch, not series:** on pins 3 and 13, the wire and the capacitor both plug into the pin's column separately. The capacitor is an extra path to GND that soaks up noise.
- **Series:** on pin 14, the 3.3 V has to pass *through* the resistor to reach the pin. Never wire 3V3 straight to pin 14.
- **Pin 16 → pin 3:** pin 16 is top-left, pin 3 is bottom row, third from left. Bend the capacitor across the gap, or put it in an empty top column and run a short wire to pin 3.
- **Why pin 16 uses 100 nF, not 10 nF:** it only has to be big enough. A larger value just slows the reference amplifier, and the 3.3 V reference never changes. Don't use 22 pF or 150 pF here — too small.
- **Pin 15:** if you have only one 3.3 kΩ, use it on pin 14 and put anything close on pin 15 (3 kΩ, 4.7 kΩ). It only cancels a tiny error.

5 · TL074 — ALL 14 PINS



Pin	Name	Connect to	Role
1	OUT A	3 kΩ to pin 2 · **test point**	Amp A output
2	−IN A	DAC0800 pin 4 (short wire)	Amp A input
3	+IN A	GND rail	Amp A reference
4	V+	Top red rail (+5V) + 100 nF to GND	Power — wire runs **up**
5	+IN B	GND rail	Amp B parked
6	−IN B	Short wire to pin 7	Amp B parked
7	OUT B	Short wire to pin 6	Amp B parked
8	OUT C	Short wire to pin 9	Amp C parked
9	−IN C	Short wire to pin 8	Amp C parked
10	+IN C	GND rail	Amp C parked
11	V−	Bottom red rail (−9V) + 100 nF to GND	Power — wire runs **down**
12	+IN D	GND rail	Amp D parked
13	−IN D	Short wire to pin 14	Amp D parked
14	OUT D	Short wire to pin 13	Amp D parked

Pin 1 — the output and test point

```
pin 1 column
● chip leg
● 3 kΩ leg ──────────► pin 2 column          (feedback: turns DAC current into voltage)
● 22 pF leg ─────────► pin 2 column          (optional: cleaner output)
○ leave EMPTY  ◄── meter red probe goes here
○ spare        ◄── hour 19: first 1.5 kΩ of the filter
```

"Test point" means the place you measure. Nothing connects to it — you leave a free hole for the probe (or push in a short jumper and touch the probe to that). Without the 3 kΩ feedback resistor the output shoots to maximum and stays there.

Feedback resistor — if you don't have 3 kΩ

Output at full scale ≈ 1 mA × the resistor. The TL074 can only reach about 3.5 V on a +5 V supply, so keep the resistor between 1.5 kΩ and 3 kΩ.

Resistor, pin 1 ↔ pin 2	Gate 2 readings (0x00 / 0x80 / 0xFF)	OK?
3 kΩ (default in this PDF)	0 / 1.5 / 3.0 V	Best
2.7 kΩ	0 / 1.35 / 2.7 V	Fine
2.2 kΩ	0 / 1.1 / 2.2 V	Fine
2 kΩ (1 kΩ + 1 kΩ in series)	0 / 1.0 / 2.0 V	Fine
1.5 kΩ	0 / 0.75 / 1.5 V	Works, smaller signal
3.3 kΩ	0 / 1.65 / 3.3 V	Top may clip
4.7 kΩ	would need 4.7 V	No — clips

Two resistors in series: first resistor from pin 1's column to an empty column, second from that same column to pin 2's column. The middle column must hold **only** the two resistor legs.

6 · CHECKS BEFORE POWER, THEN GATE 2

Beep mode — battery unclipped, USB unplugged

Probe between	Must	If wrong
Top blue ↔ bottom blue	Beep	GND rails not joined
+5V rail ↔ GND	**Not** beep	Short on the +5V side
−9V rail ↔ GND	**Not** beep	Short on the −9V side
TL074 pin 4 ↔ pin 11	**Not** beep	+5V shorted to −9V — find it before powering
DAC pins 1, 2 ↔ GND	Beep	Missing GND wire
TL074 pins 3, 5, 10, 12 ↔ GND	Beep	Missing GND wire
DAC pin 5 ↔ 6, 6 ↔ 7 … 11 ↔ 12	**Not** beep	Data wires bridged

Power up in this order

- Replace the multimeter battery first — a weak one reads about 35% high.
- Clip on the battery. DAC pin 3 and TL074 pin 11 must read **about −9 V, with the minus sign**. +9 V means the clip is reversed — fix before USB.
- Plug in USB. The top red rail reads about +5 V.
- Flash the DC test code (below). Red probe in TL074 pin 1's empty hole, black probe on GND.

```
/* USER CODE BEGIN 2 */
HAL_TIM_PWM_Start(&htim1, TIM_CHANNEL_1);
static const uint8_t test[3] = { 0x00, 0x80, 0xFF };
/* USER CODE END 2 */

  /* USER CODE BEGIN 3 */
  for (int i = 0; i < 3; i++)
  {
    GPIOE->ODR = test[i];
    BSP_LED_Toggle(LED_GREEN);
    HAL_Delay(3000);
  }
}
/* USER CODE END 3 */
```

GATE 2 — with the 3 kΩ feedback resistor

TL074 pin 1 steps every 3 seconds: **~0 V → ~1.5 V → ~3.0 V**.

The middle reading must be **half** the top one. A few tens of mV, or a tiny negative reading, at the bottom is normal.

Using a different feedback resistor? Read your numbers from the table in section 5.

Symptom	Likely cause
Always 0 V	Battery off, 3.3 kΩ on pin 14 missing, 3V3 wire missing, or DAC pin 4 not at TL074 pin 2
Stuck near +3.5 V	Feedback resistor missing, or DAC pins 2 and 4 swapped
Steps in the wrong proportions	Data wires out of order — PE7 → pin 5 … PE0 → pin 12
Reading jumps around	Loose wire — press everything in, wiggle to find it
A chip gets hot	Unplug now. Check battery polarity and that +5V and −9V aren't shorted

7 · HOUR 19 — FILTER (AMP B) AND BUFFER (AMP C)

Only after Gates 2 and 3. Remove the parking wires on amps B and C first; leave amp D parked.

From	Part	To
TL074 pin 1 (spare hole)	1.5 kΩ	Empty column X
Column X	1.5 kΩ	TL074 pin 5
Column X	150 pF (151)	TL074 pin 7
TL074 pin 5	150 pF (151)	GND rail
TL074 pin 6	Wire	TL074 pin 7 (amp B follower)
TL074 pin 7	Wire	TL074 pin 10
TL074 pin 9	Wire	TL074 pin 8 (amp C follower)
TL074 pin 8	51 Ω	Scope probe / BNC
TL074 pin 8	1 kΩ	Buzzer + (buzzer − to GND)

Cut-off about 700 kHz — it smooths the DAC steps in the ultrasonic band and changes nothing you can hear. No spare 1.5 kΩ? Two 1 kΩ work too (cut-off about 1 MHz).

What frequency to expect

Sample rate (Counter Period)	Clean tone	Notes
1 MSps (159)	~100–250 kHz	10 steps per cycle at 100 kHz — filter smooths it
2 MSps (79)	~200–500 kHz	20 steps per cycle at 100 kHz
Higher	Measure in step 9.3	DAC0800 settles in ~100 ns; TL074 is fine to ~500 kHz

> **You need a scope above 20 kHz.** The meter can't show it and the buzzer can't play it. Use `AUDIBLE 1` (2–17 kHz) for the buzzer and video, `AUDIBLE 0` (100–250 kHz) for the scope photos.

The four things that decide this

1	**Battery polarity.** −9 V on DAC pin 3 before USB goes in.
2	**Pin 4 ↔ pin 11 on the TL074 never beeps.**
3	**PE7 → pin 5.** The MSB is pin 5, the LSB is pin 12.
4	**Gate 2 passes before any DMA code.**